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# Uncovering the influence of metallic and non-metallic impurities on the ideal shear strength and ductility of Ti: An ab-initio study

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ABSTRACT

The mechanical properties of Ti alloys are significantly affected by the introduction of solute or impurity elements. Thus, the role of impurities on the hardening or softening behavior of α-Ti was investigated through first principles calculations. In particularly, to provide a comprehensive electronic and atomic basis of solute addition in α-Ti, the effect of metallic (V and Al) and non-metallic (C and O) impurities on the ideal shear strength (ISS) and generalized stacking fault energy (GSFE) across different slip systems of Ti were probed. The results revealed that the addition of V atom reduces both ISS and unstable stacking fault energy across various slip systems of α-Ti, whereas Al addition increases the ISS of Ti. Further, the addition of O atom decreases ISS for most of the slip systems while C solute atom increases ISS across all slip systems of α-Ti. To illustrate the underlying factors influencing the observed softening/strengthening behavior, the electronic density of states and valence charge transfer were determined. The electronic density of states calculations showed that the contribution from the *d* states of V atom decreases the stability of Ti-V solid solution, thereby leading to a decrease in the plastic anisotropy of α-Ti. Finally, the shearability parameter and critical resolved shear stress (CRSS) ratios across different slip systems of Ti solid solutions were calculated to understand the macroscopic effects of impurity addition on the deformation behavior of α-Ti at ambient conditions. Overall, the first-principles study provides an insight into the electronic basis of strengthening/softening effect of solute addition in α-Ti and assesses their implications on the deformation behavior of α-Ti alloys.

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## 1. Introduction

Titanium (Ti) alloys are widely used for structural applications in automotive, aviation and biomedical industries due to their high temperature stability, excellent corrosion resistance and high strength-to-weight ratio [1–3]. However, the mechanical properties of Ti alloys are significantly affected by the introduction of solute or impurity elements, such as oxygen, carbon, hydrogen etc. [4,5]. For instance, at elevated temperatures, the oxide layer grows at the Ti surface and oxygen rapidly diffuses into the metal [6,7] which leads to embrittlement and hardening of α-Ti making the deformation even more difficult [8,9]. Further, the interstitial addition of carbon also reduces the plasticity of Ti by splitting perfect dislocations into partials, which hinders the dislocation climb or cross slip [10–12]. On the other hand, substitutional solutes like aluminum, tin, zirconium, vanadium etc. are also used to improve the properties of Ti for various applications [13–15]. In Ti-Al solid solution alloys, the addition of Al led to a decrease in the critical resolved shear stress of prismatic and pyramidal ⟨c+a⟩, thereby increasing their slip activity [16]. The above mentioned experimental findings have motivated the present work, which aimed at understanding the role of solutes on the nature of plastic anisotropy and associated mechanisms in α-Ti.

In conjunction with experiments, first-principles methods are increasingly implemented to investigate the effect of different solutes on the fundamental plasticity mechanisms in α-Ti. For example, first-principles based generalized stacking fault energy (GSFE) calculations revealed that the presence of oxygen increased unstable stacking fault energy thereby increasing the Peierls stress for dislocation on the prismatic slip system of Ti [17]. First-principles methods were also used to model screw dislocation core structure and study plastic deformation in α-Ti [18–20]. Further, strong repulsive interactions were reported between interstitial oxygen and screw dislocation core on the prismatic slip

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system of α-Ti which led to solid solution strengthening by solute interaction with screw dislocations [21,22]. However, the presence of solute atom affects the equilibrium core structure and increases the complexity and accuracy of this approach. Therefore, in this work, GSFE calculations along with ideal shear strength (ISS) calculations are used to determine the active slip systems and elucidate the deformation mechanism in Ti alloys.

In this work, the stress and energy barriers for dislocation nucleation and motion in a defect free α-Ti doped with common impurity elements were estimated to provide critical information about the deformation behavior of Ti-X (X = V, Al, C and O) solid solutions. To this end, first-principles calculations were performed to quantify the effect of metallic (V and Al) and non-metallic (C and O) solutes on the ideal shear strength (ISS,τ) and generalized stacking fault energy (GSFE, γ) across different slip systems of α-Ti. The changes in electronic structure of Ti up on addition of various solute atoms were revealed through the electronic density of states (DOS) and valence charge transfer calculations. In general, the effect of different solutes on various properties of α-Ti depends on the bonding between Ti and solute atoms, which can be understood by examining the electronic structure of the solid solutions. Furthermore, shearability parameter was used to determine the strengthening or softening effect of solute atoms on different slip systems of α-Ti. Finally, a temperature dependent phenomenological model [23,24] was used to calculate the restoring stress, Peierls stress and room temperature critical resolved shear stress (CRSS) across different slip systems to understand the macroscopic effects of solute addition on the deformation behavior of α-Ti alloys.

## 2. Computational details

In the current work, ideal shear strength and stacking fault energy were calculated using Vienna Ab-initio Simulation Package (VASP) based on the density functional theory (DFT) [25,26]. Projector augmented wave (PAW) [27,28] pseudopotentials were used to represent the nuclei and core electrons of Ti and various solute atoms, with 3d and 4s as valence electrons for Ti. Exchange and correlation were treated with generalized gradient approximation (GGA) using the Perdew-Burke-Ernzerhof (PBE) formulation [29] with a cut-off energy of 350 eV for Ti. Brillouin-zone integrations were performed using a gamma pack k-point mesh of 27 × 15 × 17, selected after extensive k-point sampling convergence studies. The quasi-Newtonian method [30] was used to relax the ions with a force and energy convergence criteria of 1 meVÅ$^{-1}$ and $10^{-6}$ eV, respectively. Calculated equilibrium structure parameters such as lattice constants, cohesive energy and elastic constants for α-Ti were all found to be within 5% of the range of reported experimental and theoretical values [31,32]. The Ti-X solid solution supercell (48 atoms) for ideal shear strength calculations was obtained from 3 × 2 × 2 periodic replications of pure α-Ti unit cell (4 atoms) along $[11\bar{2}0]$, $[10\bar{1}0]$ and [0001]. Al and V solute atoms were respectively substituted for a Ti atom while, C and O atoms were respectively added interstitially at an octahedral site to obtain a solute concentration of 2 at.%. In the present study, low global concentration of V solute atom is used to obtain a stable α-Ti solid solution because the metastable β-Ti phase becomes energetically more favorable than the α-Ti phase when the concentration of V becomes large enough [33,34]. The solute atoms were added on the shear plane and were allowed to relax in all directions. The self-interactions between the solute atoms within the chosen supercell were negligible. For instance, the binding energy of Al in Ti for 48 and 72 atom supercells was found to be 0.955eV/atom. Gamma pack k-point mesh of 6 × 5 × 6 was selected, based on the convergence studies, for the basal $\{0001\}\langle 10\bar{1}0\rangle$, prismatic $\{10\bar{1}0\}\langle 11\bar{2}0\rangle$ and pyramidal $\{10\bar{1}1\}\langle 11\bar{2}0\rangle$ slip systems, whereas k-point mesh of 6 × 9 × 2 was used for the simulation cell corresponding to pyramidal ⟨c+a⟩ $\{11\bar{2}2\}\langle 11\bar{2}3\rangle$ slip system. A simple shear strain with increments of 0.01 was applied to the simulation cell along the burger's vector direction to perform the ideal shear strength calculations. Atomic positions and the atomic basis vectors were fully relaxed at each step while keeping the applied strain fixed to obtain the corresponding relaxed stresses. Periodic boundary conditions were maintained during shear deformation of the simulation cell so that the ideal shear strength calculations are independent of the position of the solute atom in the simulation cell.

Next, for the GSFE calculations, pure α-Ti unit cell was periodically replicated 2 × 2 × 3 along $[11\bar{2}0]$, $[10\bar{1}0]$ and [0001] to obtain Ti solid solution super cell (48 atoms). C and O atoms were respectively added at an octahedral site interstitially while V and Al atoms were respectively substituted for a Ti atom on the shear plane in the solid solution. The simulation setup for GSFE calculations consisted of 24 atomic layers (2 atoms in each plane) with 10 Å of free space while maintaining periodic boundary conditions. The number of atomic planes and vacuum region considered were sufficiently large to minimize any interactions between the free surfaces and slip planes. The k-point mesh density for the various solutes was selected based on convergence studies and can be found in Table S1 (refer supplementary document). The GSFE curves were calculated by displacing the desired slip plane along the Burger's vector as explained by Bhatia et al. [17,35–37]. The upper half of the crystal was shifted incrementally with linear steps along the slip direction and allowed to relax at each step. The solute atom was free to relax in all directions, whereas Ti atoms were allowed to relax only along the direction normal to the slip plane during the GSFE calculations. The changes in electronic structure of α-Ti upon addition of various solute atoms were explored through electronic density of states and valence charge transfer calculations. VESTA (visualization for electronic and structural analysis) software package [38] was used to the obtain charge density contours from the first-principles calculations.

## 3. Results and discussion

The plastic behavior of α-Ti undergoes a change upon introduction of solute atoms by either reducing or increasing the plastic anisotropy based on the chemical and elastic interactions between the solute atom and the host lattice [39,40]. To analyze the effects of metallic (V and Al) and non-metallic (C and O) solute atoms on Ti, the ideal shear strength and generalized stacking fault energy across different slip systems were measured to understand their effect on the mechanical behavior (especially dislocation nucleation and motion) of solute strengthened Ti alloys. Furthermore, their electronic structures were investigated to determine the applicability of the solute atoms as effective means of solid solution strengthening/softening in Ti. In the end, the shearability parameter and critical resolved shear stress ratios for Ti and its alloys across various slip systems were calculated to estimate the effect of solute atoms on the deformation behavior of α-Ti at ambient conditions.

### 3.1. Ideal shear strength

In the case of pure Ti, ideal shear strength of 5.18 GPa, 2.95 GPa, 3.40 GPa and 7.26 GPa were observed for the basal $\{0001\}\langle 10\bar{1}0\rangle$, prismatic $\{10\bar{1}0\}\langle 11\bar{2}0\rangle$, pyramidal $\{10\bar{1}1\}\langle 11\bar{2}0\rangle$ and

pyramidal ⟨c+a⟩ $\left\{11\bar{2}2\right\}\left\langle 11\bar{2}3\right\rangle$ slip systems, respectively, and were in agreement with previous findings [41,42] (Fig. 1). Furthermore, the prismatic slip system was found to be least resistant to the shear deformation suggesting that it would be the most active slip system during the plastic deformation of pure Ti.

In order to understand the mechanical behavior of solute strengthened Ti alloys, the shear stress-strain response of the metallic (V and Al) and non-metallic (C and O) solid solutions of α-Ti were compared to pure α-Ti along different slip systems. The substitution of V atom in Ti was found to decrease the ideal shear strength across different slip systems of Ti-V solid solution as compared to pure α-Ti (Fig. 1). For example, the addition of V in Ti decreased the ideal shear strength of the prismatic slip system from 2.95 GPa to 2.44 GPa (Fig. 1c). The prismatic slip system still has the lowest ideal shear strength; however the anisotropy between prismatic and pyramidal slip systems notably decreased which could be of vital importance to improve the room temperature formability of Ti alloys. Further, the substitution of Al solute atom in Ti increased the ideal shear strength on the prismatic, pyramidal and pyramidal ⟨c+a⟩ slip systems; whereas a small decrease in the ideal shear strength was observed on the basal slip system with respect to pure α-Ti (Fig. 1). Likewise, the interstitial addition of C and O atoms in Ti also had significant effect on the ideal shear strength of pure Ti. The addition of C atom in Ti was found to increase the ideal shear strength on the basal, prismatic and pyramidal slip systems; while the ideal shear strength decreased for the pyramidal ⟨c+a⟩ slip system as compared to pure α-Ti (Fig. 1). For instance, the addition of C solute atom in Ti increased the ideal shear strength of the prismatic slip system from 2.95 GPa to 3.36 GPa (Table S2, refer supplementary document). On the other hand, upon addition of O solute atom in Ti an infinitesimal decrease in the ideal shear strength was observed for the prismatic, pyramidal and pyramidal ⟨c+a⟩ slip systems when compared to pure α-Ti (Fig. 1).

### 3.2. Electronic structure

#### 3.2.1. Electronic density of states (DOS)

The electronic DOS for the various solute strengthened Ti alloys were obtained for the unstrained configurations (Fig. 2). The region below the Fermi level (0 eV) corresponds to the occupied bonding states (ground states); likewise, the region above the Fermi level corresponds to the unoccupied antibonding states (excited states). The substitution of V atom increased the DOS near the Fermi level due to the overlap between the non-symmetric *3d* states of Ti and V leading to a decrease in the electronic stability of Ti-V alloys as compared to pure α-Ti (Fig. 2). On the other hand, addition of Al solute atom was found to have negligible effects on the DOS of Ti alloys (Fig. 2). Addition of both C and O solute atoms decreased the DOS near the fermi level due to the presence of symmetric *s* and *p* states and increased the electronic stability of the solute strengthened Ti alloy (Fig. 2). Overall, the addition of V atom decreased the electronic stability of Ti thereby decreasing the bond strength between the solute and host atoms however; addition of Al, C and O solute atoms had very small effect on the structural stability of Ti alloys, respectively.

#### 3.2.2. Valence charge transfer

The nature of atomic bonding determines the mechanical response of a material; thus, the valence charge density distributions were evaluated to provide qualitative insights into the effects of addition of various solute atoms [43]. For instance, anisotropic distribution of charge around an atom indicates an increase in the

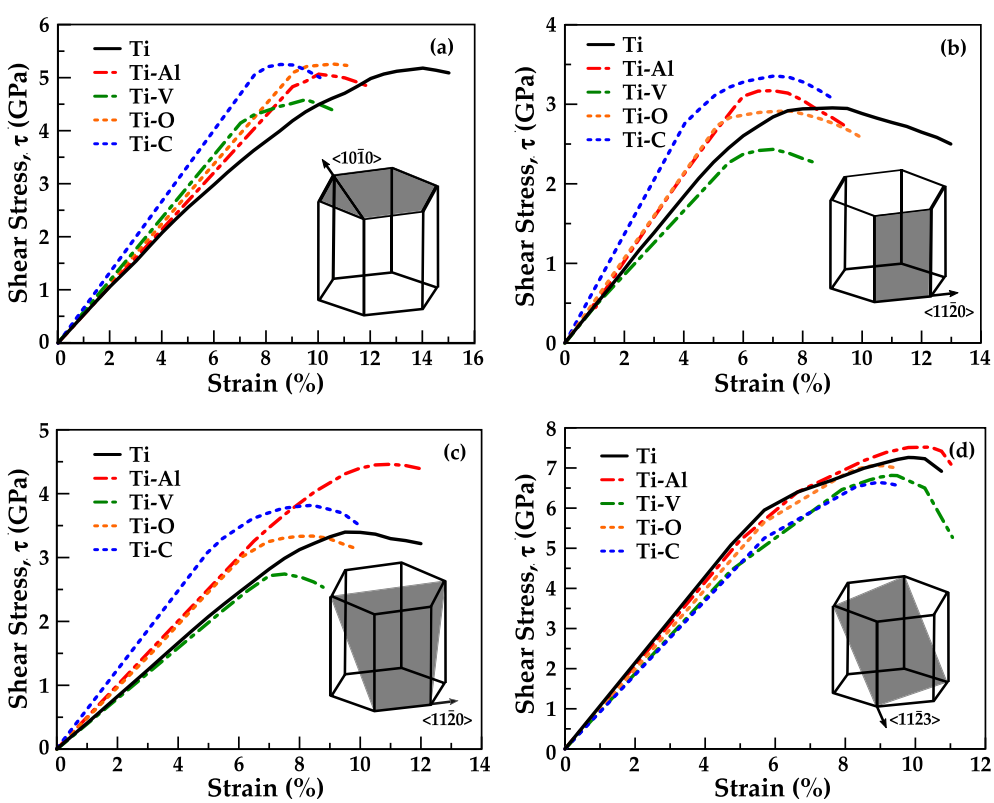


**Fig. 1.** The shear stress–strain response of the (a) basal $\{0001\}\langle 10\bar{1}0\rangle$ slip system, (b) prismatic $\left\{10\bar{1}0\right\}\left\langle 11\bar{2}0\right\rangle$ slip system, (c) pyramidal $\left\{10\bar{1}1\right\}\left\langle 11\bar{2}0\right\rangle$ slip system and (d) pyramidal ⟨c+a⟩ $\left\{11\bar{2}2\right\}\left\langle 11\bar{2}3\right\rangle$ slip system for metallic (V and Al) and non-metallic (O and C) solutes respectively as compared to pure α-Ti along different slip systems.

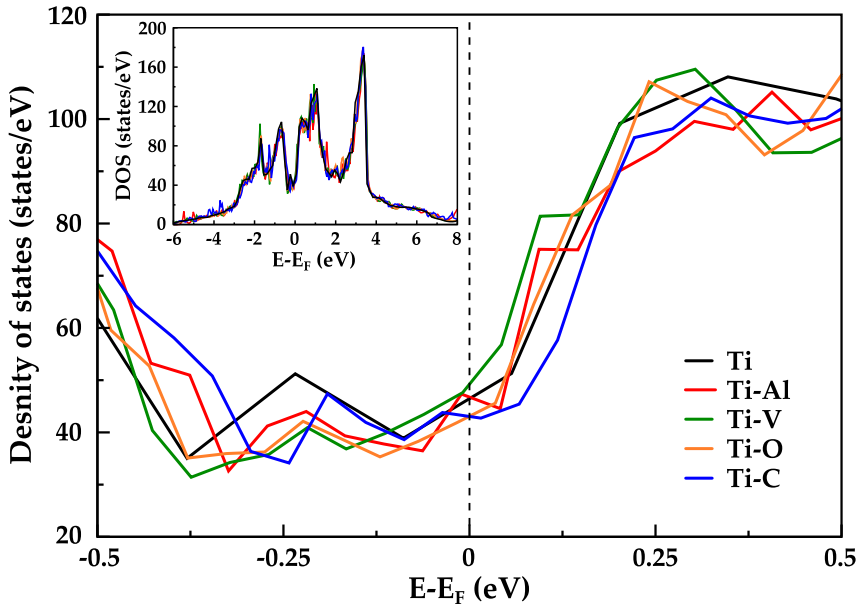


**Fig. 2.** Electronic density of states for Ti with metallic and non-metallic solutes respectively near the fermi level at 0 eV.

ability to withstand shear deformation, whereas; isotropic distribution of charge promotes shear deformation [44,45]. Therefore, evaluating the change in valence charge distribution upon solute addition can shed light on the observed changes in the deformation behavior across different slip systems of various Ti solid solutions. The valence charge density difference was calculated as:

$$\Delta\rho = \rho_{\mathrm{Ti-X}} - \rho_{\mathrm{Ti}} - \rho_{\mathrm{X}} \quad (1)$$

where $\Delta\rho$ represents the valence charge density difference, $\rho_{\mathrm{Ti-X}}$ represents the valence charge density of Ti-X solid solution, and $\rho_{\mathrm{Ti}}$ and $\rho_{\mathrm{X}}$ represent the valence charge density of non-interacting Ti atoms and an X atom, respectively [46–48]. Shearability of the solid solution is mainly affected by the bonding between Ti and solute atoms as the charge transfer takes place around the solute atom. Fig. 3 shows the planar charge density on the 1) basal, 2) prismatic, 3) pyramidal and 4) pyramidal ⟨c+a⟩ slip systems. In the charge density transfer contour, the solute atom is represented by a white circle in the center and surrounded by the Ti atoms. The charge distribution is highly isotropic around each of the solute atoms on the prismatic slip system which explains the lowest ideal shear strength of the prismatic slip system across various slip systems of Ti alloys. In the case of metallic solute atoms, the valence charge transferred from V and Al towards Ti atoms and gets accumulated in the region between solute and Ti atoms (Fig. 3a and b). Further, the charge depletion around the solute atom is more pronounced upon substitution of V as compared to Al, hence decreasing the strength of bonds between V and Ti atoms thereby explaining the lower ISS across each slip system of Ti-V as compared to Ti-Al solid solution. However, up on addition of C and O atoms valence charge is accumulated around the non-metallic solute atoms (Fig. 3c and d). Highly anisotropic charge distribution on the pyramidal ⟨c+a⟩ slip system contributes to the bond strength as compared to the basal and prismatic slip systems with a uniform (spherical) charge density distribution.

### 3.3. Generalized stacking fault energy

In a GSFE curve, the unstable stacking fault energy ($\gamma_{usf}$) is a maxima and a measure of the energy barrier for dislocation motion [49]. Whereas, intrinsic/stable stacking fault energy ($\gamma_{isf}$) is a local minima and indicates to the existence of metastable structures formed during crystal slip, essential for dislocation dissociation [49,50]. In pure Ti, dislocation dissociation was observed on the basal, prismatic and pyramidal ⟨c+a⟩ slip systems, with prismatic slip system having the lowest stable ($\gamma_{isf} = 237\mathrm{mJ/m^2}$) and unstable ($\gamma_{usf} = 276\mathrm{mJ/m^2}$) stacking fault energies. The calculated GSFE curves of pure α-Ti across various slip systems were found to be in good agreement with previous DFT calculations [12,33,50].

To examine the effect of solute atoms on the deformation behavior of α-Ti, the stacking fault energies curves across various slip systems of Ti alloys as a function of the applied displacement along the burger's vector direction were investigated. For the basal slip system, both metallic and non-metallic solute atoms respectively decreased the stable($\gamma_{isf}$) and unstable($\gamma_{usf}$) stacking fault energies of α-Ti (Fig. 4a and b). The decrease in $\gamma_{isf}$ is more pronounced upon addition of the non-metallic impurities as compared to the metallic solute atoms. For instance, the stable stacking fault energy for the basal slip system of Ti reduced from 278.45 mJ/m$^2$ to 95.89 mJ/m$^2$ upon addition of C solute atom. The addition of V and Al solute atoms decreased $\gamma_{usf}$ on the prismatic slip system of Ti (Fig. 4c). However, the effect of metallic solute atoms on $\gamma_{isf}$ of prismatic slip system was opposite because the addition of V increased $\gamma_{isf}$ from 237 mJ/m$^2$ to 251 mJ/m$^2$ and addition of Al decreased $\gamma_{isf}$ from 237 mJ/m$^2$ to 206 mJ/m$^2$ (Table S3). On the other hand, addition of both C and O solute atoms increased $\gamma_{isf}$ on the prismatic slip system of Ti (Fig. 4d). The dislocation became narrower upon addition of O solute atom since the difference between $\gamma_{isf}$ and $\gamma_{usf}$ decreased significantly. For the pyramidal slip system of various Ti alloys, dislocation dissociation was not observed since there was only a single maximum in the GSFE curve corresponding to the unstable stacking fault energy (Fig. 4c). Addition of metallic solute atoms decreased $\gamma_{usf}$ whereas, the addition of non-metallic solute atoms increased $\gamma_{usf}$ for the pyramidal slip system of Ti alloys. For example, $\gamma_{usf}$ on the pyramidal slip system (761 mJ/m$^2$) of pure α-Ti decreased to 496 mJ/m$^2$ upon

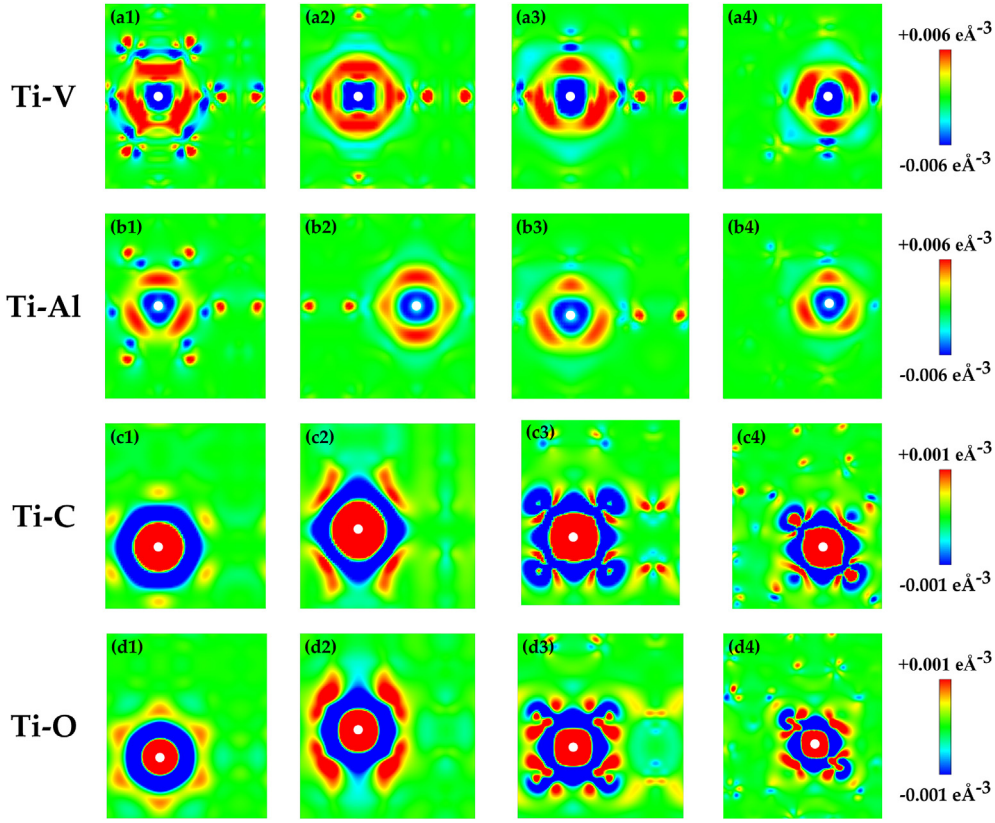


**Fig. 3.** Planar charge density (eÅ$^{-3}$) on the (1) basal, (2) prismatic, (3) pyramidal and (4) pyramidal ⟨c+a⟩ slip systems around the solute atom in (a) Ti-V, (b) Ti-Al, (c) Ti-C and (d) Ti-O solid solutions. The solute atom is represented by a white circle in the center of the charge density contours.

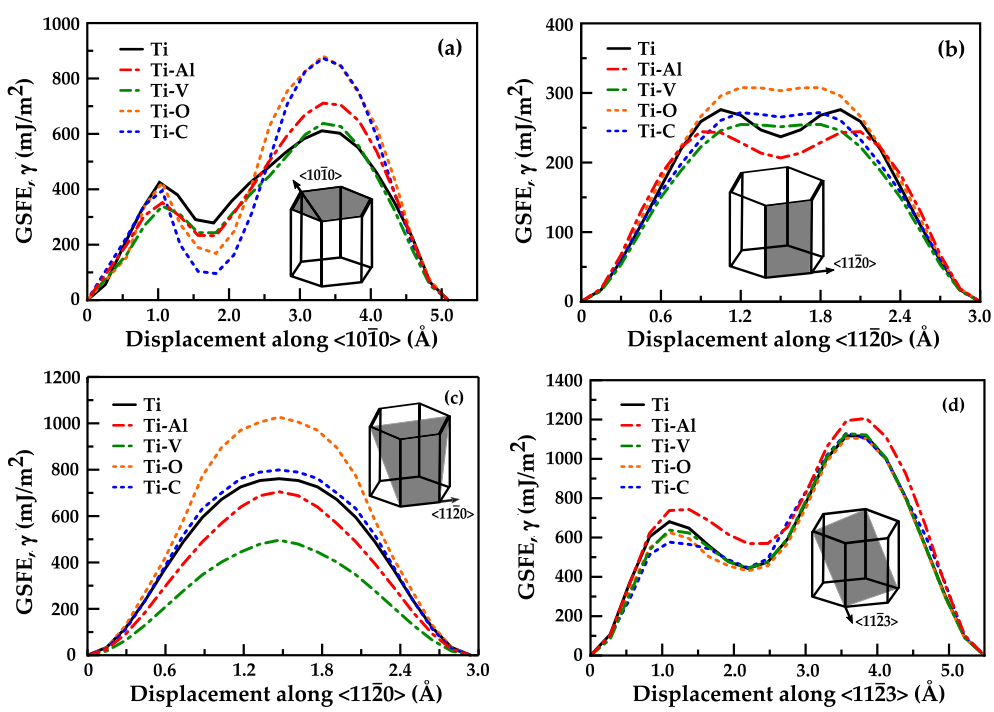


**Fig. 4.** Generalized stacking fault energy curves for the (a) basal $\{0001\}\langle 10\bar{1}0\rangle$ slip system, (b) prismatic $\{10\bar{1}0\}\langle 11\bar{2}0\rangle$ slip system, (c) pyramidal $\{10\bar{1}1\}\langle 11\bar{2}0\rangle$ slip system and (d) pyramidal ⟨c+a⟩ $\{11\bar{2}2\}\langle 11\bar{2}3\rangle$ slip system for metallic (V and Al) and non-metallic (O and C) solutes respectively as compared to pure α-Ti along different slip systems.

addition of V solute atom whereas, the addition of O atom increased $\gamma_{usf}$ to 1026 mJ/m$^2$ (Table S3, refer supplementary document). The metallic solute atom Al increased both stable and unstable stacking fault energies for the pyramidal ⟨c+a⟩ slip system when compared to pure α-Ti (Fig. 4d). Conversely, the substitution of V solute atom decreased $\gamma_{usf}$ and had negligible effect on $\gamma_{isf}$ for the pyramidal ⟨c+a⟩ slip system of Ti (Table S3). On the other end, non-metallic solute atoms decreased $\gamma_{usf}$ for the pyramidal ⟨c+a⟩ slip system

however, their effect on $\gamma_{isf}$ was very small as compared to pure α-Ti (Fig. 4d).

### 3.4. Shearability parameter

Physically, in a perfect crystal, ideal shear strength represents the stress required to shear atomic planes. To understand the microscopic plastic deformation behavior a sinusoidal relationship between shear stress and displacement was proposed [51], which was used to proposed a criterion to predict the dominant slip system in a given metal based on the GSFE and elastic properties [52]. This criterion was further extended to propose shearability parameter to determine the effect of solute atoms on the ideal shear strength and elastic properties [44]. Thus, the shearability parameter ($R_\tau$) for Ti alloys is defined as:

$$R_\tau = [\tau_{Pr}/C_{Pr}]/[\tau_P/C_P] \quad (2)$$

where, $\tau_{Pr}$ and $C_{Pr}$ are the ideal shear strength and the shear modulus along the prismatic slip system, and $\tau_P$ and $C_P$ are the ideal shear strength and shear modulus along the investigated slip system. Prismatic being the primary slip system for deformation in pure α-Ti, the shearability parameter ($R_\tau$) is an indicator of solute strengthening/softening behavior for a given slip system with respect to the primary slip system. For example, a $R_\tau$ value greater than or equal to 1 for a non-primary slip system indicates to its higher or equal activation probability as compared to the prismatic slip system in Ti alloys. In pure α-Ti as well as in Ti-X solid solutions the prismatic slip system has $R_\tau$ value of 1 which indicates that the prismatic slip system has maximum shearability (Fig. 5). The addition of metallic solutes slightly increased the shearability along pyramidal slip system as the $R_\tau$ value increased from 0.76 to 0.81 upon addition of V solute atom to Ti. Similarly, the $R_\tau$ value increased from 0.26 for pure α-Ti to 0.55 and 0.45 upon addition of C and O solute atoms respectively along the pyramidal ⟨c+a⟩ slip system, thus increasing the shearability of pyramidal ⟨c+a⟩ slip system (Table S4, refer supplementary document). These observations from the ideal shear strength calculations are found to be consistent with the experimentally and theoretically reported behavior of Ti alloys [53–55].

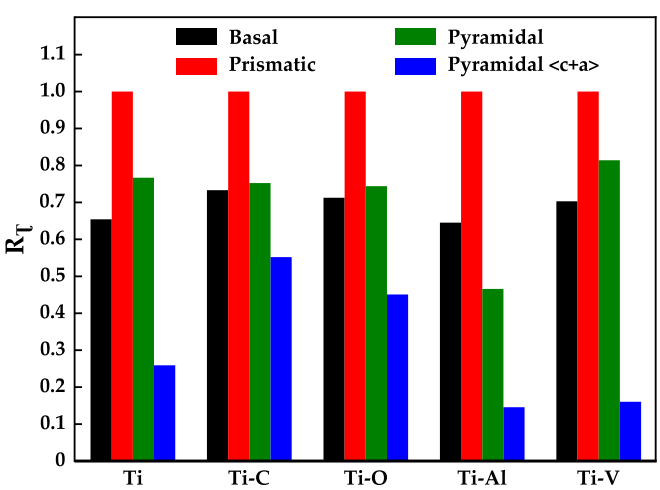


**Fig. 5.** Shearability parameter ($R_\tau$) with respect to ideal shear strength across various slip systems of Ti alloys.

### 3.5. Critical resolved shear stress (CRSS) ratios

It is well known that the critical stress required to move a stationary dislocation in a perfect crystal at 0 K is known as Peierls stress [56], however; the Peierls stress can be significantly changed by the presence of alloying elements [57]. With the development of reliable and accurate methods to deduce the atomic interactions across the slip plane using the *ab-inito* energy calculations; numerical and analytical methods have been employed to calculate the Peierls stress ($\sigma_p$) [24,58,59]:

$$\sigma_p = 2K \exp\left(-\frac{4\pi\zeta}{b}\right) \quad (3)$$

where ζ is the dislocation core half width and is defined as:

$$\zeta = \frac{Kb}{4\pi F_{\max}} \quad (4)$$

$b$ is the Burgers vector and $K$ is an energy factor ($K$ is $\mu/2\pi$ for a screw dislocation where $\mu$ is shear modulus). Effect of anisotropy in hcp Ti on the energy factor is taken into consideration using the data from Ref. [60]. The maximum restoring stress ($F_{max}$) corresponding to a disregistry vector $f(x)$ along the glide plane is defined as:

$$F_{\max} = \left|\frac{d\gamma(f(x))}{df(x)}\right|_{\max} \quad (5)$$

The restoring stress can be obtained from the gradient of the GSFE curve in the direction of the minimum energy path and then substituted back in equation (3) to calculate $\sigma_p$ using equation (2). The restoring stress from the GSFE calculations were obtained for pure α-Ti and Ti-X solid solutions across various slips systems and were found to match very well with the values available in literature [50] (Table 1).

The calculated values of restoring stress for different slip systems of each solid solution can be used to calculate the corresponding Peierls stresses as described in equations (2)–(4). The *ab-intito* data at 0 K was extrapolated to calculate Peierls stress at 300 K, i.e., CRSS using the phenomenological description proposed by Kocks et al. to evaluate the effect of solute atoms on CRSS ($\sigma_{CRSS}$ (300)) of Ti [23]:

$$\sigma_{CRSS}(T) = \sigma_p^0\left(1 - \left(\frac{T}{T_a}\right)^{1/q}\right)^{1/p} \quad (6)$$

where $\sigma_p^0$ is the Peierls stress at absolute zero temperature, $T_a$ is the

**Table 1**
The maximum restoring stress ($F_{max}$) of pure α-Ti and Ti-X solid solutions.

| Restoring Stress (GPa) | | | | | |
|---|---|---|---|---|---|
| | Ti | Ti-V | Ti-Al | Ti-C | Ti-O |
| Basal | 6.44 | 6.44 | 7.83 | 8.89 | 8.53 |
| Prism | 3.74 | 3.31 | 4.27 | 3.50 | 3.86 |
| Pyramidal | 8.94 | 5.04 | 7.25 | 9.69 | 13.07 |
| Pyramidal ⟨c+a⟩ | 9.65 | 9.28 | 10.99 | 9.96 | 9.11 |

athermal temperature, $p$ and $q$ are two material parameters that determine the temperature dependence of the Peierls stress. $T_a$ is a critical temperature above which the yield stress loses its dependence on the temperature, i.e. the thermal component of the yield stress becomes zero [23]. Note that the temperature dependence of a slip system can be a complex function of composition, tendency to cross-slip, dislocation character, etc., and Kocks' formula only provides an estimate of the CRSS value for each slip system [61,62]. A linear dependence of the thermal component of yield stress with the square root of temperature has been reported for polycrystalline and single crystal Ti with different solute concentrations (such that $p \approx 1$ and $q \approx 2$) [63–65]. In pure α-Ti as well as Ti-X solid solutions, prismatic slip system has the lowest ideal shear strength, unstable stacking fault energy as well as critical resolved shear stress. Thus, prismatic slip system is the most favorable slip system for deformation in Ti and its alloys as also observed in previous studies [63,66]. Hence, the CRSS calculated for each slip system was normalized with respect to the CRSS of the most active prismatic slip system to determine the CRSS ratio for different slip systems of each Ti solid solution.

The addition of metallic solute atoms decreased the CRSS required to activate pyramidal slip system in Ti with respect to the prismatic slip system with increase in temperature (Fig. 6a–c). Furthermore, at 300 K the CRSS ratio for the pyramidal slip system

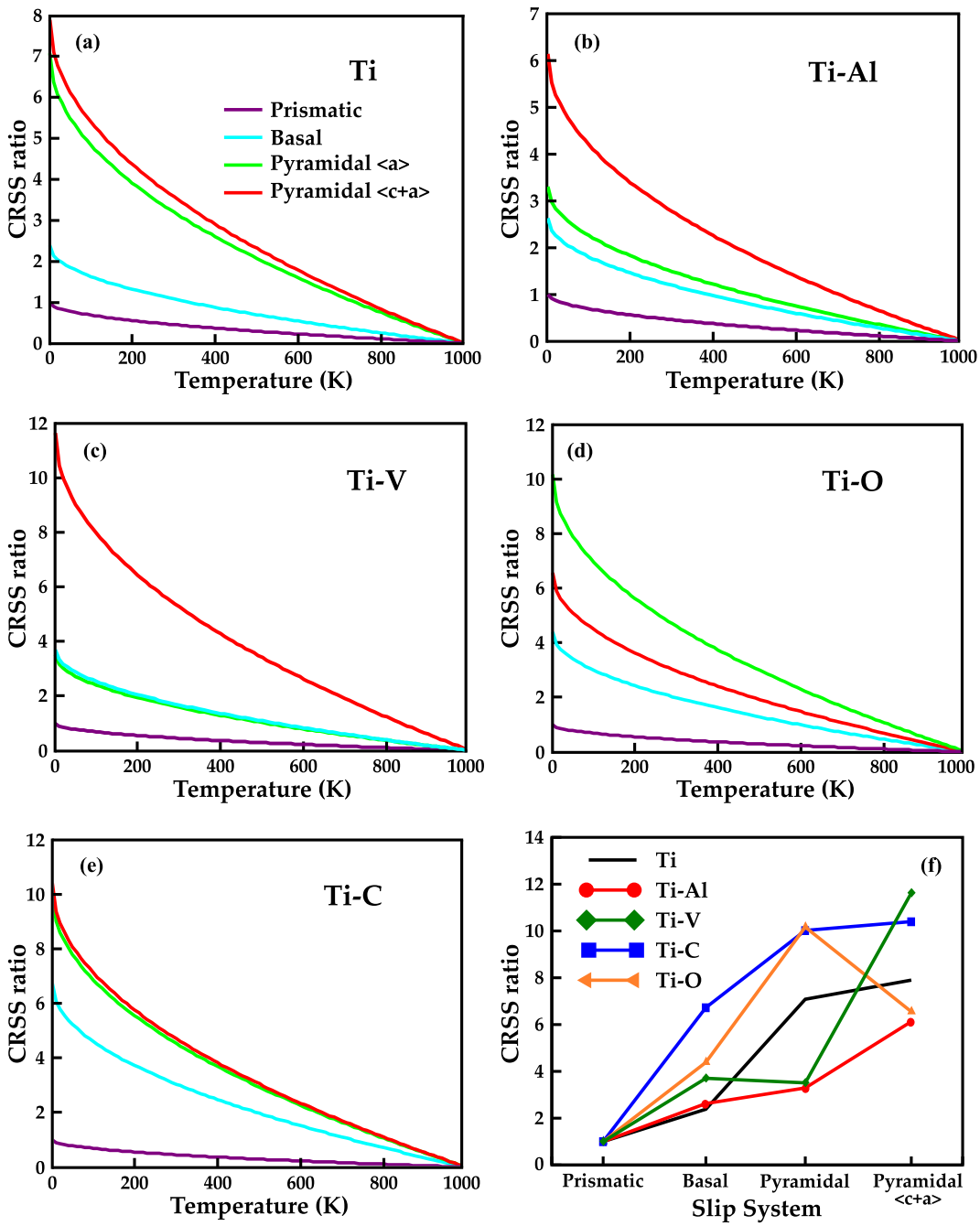


**Fig. 6.** Temperature dependence of CRSS ratios for the four studied slip systems of (a) pure α-Ti, (b) Ti-Al, (c) Ti-V, (d) Ti-O and (e) Ti-C with respect to the prismatic slip system. (f) The room temperature (300 K) CRSS ratio for pure Ti, Ti-Al, Ti-V, Ti-O and Ti-C with respect to the corresponding prismatic slip system in each case.

was found to be lower than the basal slip system in Ti-V solid solution (Fig. 6f). Pyramidal slip systems were found to coexist with basal and prismatic slip systems in Ti-6Al-4V titanium alloy, with basal and prismatic as more dominant deformation modes than the pyramidal slip systems [55]. Thus, the pyramidal slip system has a higher slip activity thereby decreasing the plastic anisotropy of Ti upon addition of V and Al solute atoms, respectively. The addition of O atom increased the CRSS ratio value of basal and pyramidal slip system but decreased the relative resistance of pyramidal ⟨c+a⟩ slip system, promoting its higher slip activity (Fig. 6d). Pyramidal ⟨c+a⟩ dislocations have been observed in various oxygen doped titanium alloys [67–69] thereby decreasing their plastic anisotropy. The interstitial addition of C atom increased the CRSS ratio across all slip systems of Ti (Fig. 6e) and thus increased the plastic anisotropy titanium alloys [10,11].

Ti alloys are highly anisotropic in the elastic as well as plastic deformation, as clearly indicated by the shearability parameter and CRSS ratios (Figs. 5 and 6). Thus, crystallographic texture plays a crucial role in the macroscopic response of polycrystalline samples. Furthermore, concentration of different solutes in α-Ti can significantly influence the slip activities, which need to be considered. Nevertheless, the CRSS ratio values can serve as an important input parameter to a mesoscale crystal plasticity model to reproduce the macroscopic response and also study the local microscopic deformation mechanisms of titanium alloys.

## 4. Conclusions

An extensive first-principles study was performed to understand the impacts of solute addition on the plastic deformation of Ti alloys. Particularly, the effect of metallic (V and Al) and non-metallic (C and O) solutes on the ideal shear strength across different slip systems of α-Ti was accurately determined. Additionally, the electronic structure calculations through density of states and valence charge transfer contours illustrated the underlying reasons for the observed shear response of various Ti binary alloys. Further, the GSFE calculations were performed to determine the energy barriers for dislocation motion. Furthermore, macroscopic effects of solute addition were explored by calculating the shearability parameter and CRSS ratios across different slip systems of Ti alloys with respect to the prismatic slip system. The key findings of this work are:

- The addition of V solute atom decreases the ideal shear strength across various slip systems of α-Ti. However, the addition of Al and C solute atoms increases the ideal shear strength across different slip systems of α-Ti with the exception of basal slip system for Al addition and pyramidal ⟨c+a⟩ slip system for C addition. Further, addition of O atom slightly decreases the ideal shear strength across different slip systems of α-Ti except basal slip system.
- The interactions of the *d* states of V with the *d* states of Ti contributed to a decrease in structural stability that led to a decrease in the ideal shear strength across various slip systems of Ti-V solid solution.
- The addition of non-metallic solute atoms increases the shearability of pyramidal ⟨c+a⟩ slip system from 0.26 to 0.55 and 0.45 respectively. On the other hand, addition of V solute atom to α-Ti slightly increases the shearability of pyramidal slip system.
- The CRSS ratios across different slip systems of Ti alloys were calculated, using generalized stacking fault energy calculations, with respect to the prismatic slip system. Addition of V and Al solute atoms increases the slip activity of pyramidal slip system and thus reducing plastic anisotropy in α-Ti at room temperature.

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## Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jallcom.2019.02.231.

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